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Assessing the perception of overall indoor environmental quality: Model validation and interpretation

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ABSTRACT

The indoor environmental quality (IEQ) was determined by assessment of its various domains, among which the four principal ones are thermal, acoustic, visual and indoor air quality (IAQ). While many IEQ models have been established to develop scheme for assessing the overall IEQ, the lack of model validation significantly limited their applicability. This study aimed to provide new insights into how occupants evaluate the overall IEQ performance by establishing regression models and machine learning models that predict occupants' overall satisfaction using their satisfactions with the four principal domains of IEQ. Accuracy and applicability of proposed models were examined on two different datasets. Prediction error increased 21% when models were generalized to new dataset collected from different buildings and populations, and error of machine learning models increased more significantly than regression models. The model performance varied across building types. Models trained on samples collected from office settings had better generalization to the similar environment, while increased error was identified for schools and residences. The acoustic environment had a greater impact on females than males, while the IAQ had a greater impact on males. The Shapley Additive Explanations was applied to interpret model predictions and measure contribution of the variables. The results revealed that the unsatisfying IEQ factor often yielded a dominant negative impact on the overall IEQ satisfaction, which can hardly be compensated by a higher satisfaction with another factor. Efficient strategy can be developed to improve overall satisfaction with IEQ based on findings of the present study.

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1. Introduction

Sustainable buildings aim to achieve high efficiency in use of energy, materials and space, while providing satisfying indoor environmental quality (IEQ) to occupants [1]. As humans spend about 90% of their time indoors [2], overwhelming evidences demonstrated that IEQ has a significant impact on occupants' health, well-being and productivity [3,4]. Researchers have been committed to exploring the links between various IEQ factors and environmental perceptions in an effort to create a healthy, comfortable and productivity IEQ, which is the primary goal for building facilities management [5].

Generally, the standards for IEQ evaluation did not directly assess the overall indoor environment, but rather assess the aspects of indoor environment separately, with the most consideration given to thermal, acoustic, visual environment and indoor air quality (IAQ) [6]. These factors were also identified as the principal factors for characterizing the acceptability of IEQ by ASHRAE standard [7]. Other than the four principal factors, occupants' perception was also considered to be affected by several additional factors including occupant density, privacy, layout, views to outdoors, etc. It is also documented that, to some degree, people integrate their responses to individual IEQ factors when evaluating the overall performance of IEQ. Therefore, the assessment of the different domains of IEQ was often aggregated under a scheme as the assessment of the overall environment [8]. A common way to establish such scheme is by interpreting quantitative relationships between occupants' satisfaction with each environmental factors (single-domain satisfaction) and their satisfaction with the overall indoor environment (overall satisfaction) [9].

Table 1 presents a summarize of studies where relationship between occupants single-domain satisfactions and the overall sat-

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Nomenclature

IEQ	Indoor environmental quality
IAQ	Indoor air quality
ML	Machine learning
SHAP	Shapley additive explanations
TS	Thermal satisfaction
AS	Acoustic satisfaction
VS	Visual satisfaction
IAQS	Indoor air quality satisfaction
ANOVA	Analysis of variance
PCA	Principle component analysis
WGM	Weighted geometric mean
WHM	Weighted harmonic mean
RF	Random forest
Xgboost	Extreme gradient boost
CV	Cross validation
RMSE	Root mean squared error
MAE	Mean absolute error
MAPE	Mean absolute percentage error
NLS	Nonlinear least squares
OLS	Ordinary least squares

isfaction was explored. The major findings of those studies clarified the contribution of individual environmental factors to occupants perceived overall indoor environment. Different IEQ assessment models have been developed based on survey data [8,10–18], and the linear and logistic regression analysis were the most adopted approach. Although the thermal, acoustic, visual and IAQ satisfactions were considered as the primary factors of IEQ, not all studies included all of them in their models. The studies by Mui and Chan [10] did not include visual satisfaction and the study by Buratti et al.[16] did not include IAQ satisfaction. In the studies by Kim and de Dear [19] and Cheung et al. [20], the secondary factors from each primary factors, for example, noise level and sound privacy from acoustic, were used. Two studies [15,16] used a linear weighting scheme but the coefficients of variables were not estimated by regression analysis. Instead, the respondents were asked to self-report their evaluations of importance for each environmental factor. Wong et al., developed an open acceptance model with more flexibility based on frequency distribution functions and compared it with the logistic model that they previously proposed [17]. Tang et al. tested non-linear regression model against the linear model and found that the non-linear model had higher accuracy [18].

Attempts were also made to interpret the overall satisfaction using other explanatory models and statistical analysis. Based on occupant survey database from Center for the Built Environment, Kim and de Dear adapted the Kano's satisfaction model, which was originally developed for marketing, to identify the Basic Factors and the Proportional Factors in the context of IEQ [19]. In a laboratory study, Yang and Moon used the factorial analysis of variance (ANOVA) to measure effects of environmental factors and found the acoustics had the greatest impact on the overall satisfaction [21]. Cheung et al. trained and compared three linear models – linear regression model, mixed effect linear regression model, and linear regression model using variables identified by principle component analysis (PCA) [20]. They found that some of the secondary factors did not have significant impact on the overall satisfaction except when they merged into a major factor using PCA, e.g., stuffiness and odors merged into perceived air quality. Wei et al. reviewed the credits assigned to environmental factors in the green building standards for the assessment of IEQ, and reported that IAQ was given the highest credits, followed by thermal, visual and acoustic environment [6].

Although reported to be valid for the data obtained, there is still concerns with applicability of the proposed IEQ assessment models in the literature. First, the lack of model comparison is a critical obstacle to evaluating the accuracy and robustness of the proposed models. Since most of the previous studies presented only a single model, there is a great need for model comparison analysis to examine the performance of different statistical models, as well as other artificial intelligence approaches such as machine learning, which have been widely used in predicting IEQ and interpreting occupants' perception and comfort [22–24]. Second, the proposed IEQ assessment models lacked validation on separate data that was not used in construction of the models, and consequently it is not clear whether the models were overfitted and may fail to fit additional data reliably [25]. This concern is particularly serious because of the identified inconsistency between the proposed models (discussed in subsection 6.4). The fact that most IEQ models were trained using data collected from uncontrolled environment also added to the concern that the model could learn too much from a noisy dataset, in which case response of occupants could be influenced by a complex set of factors which cannot be accounted for solely by physical measurements [26]. In addition, the limitation of population was not properly addressed

Table 1
Summarize of studies where IEQ models were established to interpret occupants' overall satisfaction.

Studies	Controlled or uncontrolled environment	Environment setting	Method
Mui & Chan, 2005 [10]	Uncontrolled	Office	Linear regression
Wong et al., 2008 [11]	Uncontrolled	Office	Logistic regression
Cao et al., 2012 [12]	Uncontrolled	Office, teaching, libraries	Linear regression
Ncube & Riffat, 2012 [13]	Uncontrolled	Office	Linear regression
Kim & de Dear, 2012 [19]	Uncontrolled	Office	Kano's model
Heinzerling et al., 2013 [8]	Uncontrolled	Office	Linear regression
Fassio et al., 2014 [14]	Uncontrolled	University	Linear regression, logistic regression
Mihai & Iordache, 2016 [15]	Uncontrolled	University	Linear relationship with self-reported weightings
Buratti et al., 2018 [16]	Uncontrolled	University	Linear relationship with self-reported weightings
Wong et al., 2018 [17]	Uncontrolled	Office, residence, classrooms	Frequency distribution functions
Yang & Moon, 2019 [21]	Controlled	University	ANOVA
Tang et al., 2020 [18]	Controlled	Office	Linear regression, non-linear regression
Cheung et al., 2021 [20]	Uncontrolled	Office	Linear regression, linear mixed effect model

and discussed in literature. Individual differences in demands and preferences for IEQ have been recognized as a key driver of low occupant satisfaction [27]. Previous studies indicated that such individual difference was likely to be caused by factors such as building type [28], gender [29], age [30] and climatic background [31]. Most of the studies focused only on educational buildings or office buildings (Table 1), with the university student and office staff being the most interviewed respondents. It is unclear that whether a IEQ model fitted on datasets collected from school and office can be applied interchangeably or generalized to other indoor environment such as residential buildings. Increased error is expected during generalization beyond the subject samples in the training dataset [32], but we still have very little empirical evidence with such application. Therefore, the outcomes of this study are expected to answer the following questions:

1. What is the appropriate model for explaining relationship between occupants' satisfaction with individual environmental factors and the overall environment?
2. When an IEQ assessment model fitted to a dataset is applied to additional data collected from different populations and environment, how much would the error increase and what factors contribute to the error?
3. What inferences can be drawn from the established models and how these inferences will affect the management of IEQ?

2. Research outline

In response to the unexplored issues mentioned above, this study developed, verified, and compared different models on two different datasets to enhance the existing understanding of relationships between individual factors and the overall environment by improving the accuracy of the IEQ assessment model and providing additional model explanation information. The schematic outline of the study is illustrated in Fig. 1. Two datasets collected from different indoor environment and populations were investigated in this study: dataset I was collected from controlled exposure experiments with recruited university students and dataset II was collected from existing buildings through a nationwide online survey. Three regression models – linear model, weighted geometric mean (WGM) model and weighted harmonic mean (WHM) model, and two machine learning (ML) models – random forest (RF) and Xgboost, were employed to predict the overall satisfaction using the thermal, acoustic, visual and IAQ satisfactions (see subsection 4.1 and 4.2 for model information). The performance of the proposed models was evaluated with a double validation on the two datasets: First, the prediction error was estimated within the dataset I using a 10-fold cross validation. Then, the models trained on dataset I was tested on dataset II to evaluate the applicability of models on external data. At last, to extract more insights from the proposed models regarding how the overall satisfaction was impacted by the single-domain satisfactions, we used a unified framework – SHAP (SHapley Additive exPlanation) to interpret resulting models' predictions by measuring contribution of each single-domain satisfaction with SHAP values.

3. Characteristic of datasets

3.1. Overview

Two datasets collected from different environment and populations were adopted to evaluate performance of the proposed models in this study. Characteristics of the two datasets are presented in Table 2. Dataset I consists of two experiments where a total of 56 recruited subjects were exposed to 43 controlled IEQ conditions in a lab configured to emulate office setting [18,80]. Dataset II was collected by conducting a nationwide online survey with 1869 respondents from various of buildings. Different from that all subjects in dataset I were university students, 86% of respondents in dataset II were manager, engineer, administrator, salesperson, and worked in other occupations. There were also more variations in respondents age in dataset II (see Fig. 6 for age range distribution). Most of the respondents in dataset II were surveyed when they were self-reported to be in office buildings, residential buildings and school, while a very small number of them were in hospital, shopping mall and hotel. Sixteen percent of respondents reported that they were outdoors or in a vehicle.

3.2. Dataset I – Controlled exposure experiment

In the controlled exposure experiments, all the subjects were university students with age between 20 and 25 and each of them only participated one of the experiments. Subjects followed similar instructions in the two experiments. They were asked to be seated and do normal office work such as reading and typing. Uniform dress code is required in both experiments to remain a clothing insulation of 0.5 (experiment I) and 0.57 (experiment II). Both experiments were conducted in summer in Chongqing, China. Subjects' satisfactions with thermal, acoustic, visual, IAQ and overall environment were measured with a paper copy of questionnaire during the exposure. The exposure durations were 3 h and 1 h in the experiment I and II, respectively. It is noted that different parameters of thermal, acoustic, visual and IAQ were simulated, and different levels of parameters were used in the two experiments, as shown in Table 3. These levels were designated and combined using experimental design methods including orthogonal design and uniform design [33]. More details of the experiments, i.e., the number of votes made by each subject and the complete sets of IEQ conditions in the two experiments, were presented in Supplementary Material. The experiments were approved by the Ethics Review Committee for Life Sciences Study of Central China Normal University with a registration number: CCNU-IRB-2018-012.

3.3. Dataset II – Online survey

The online survey was launched from May 2021 to October 2021 to obtain evaluations of IEQ in various buildings in China. The survey consists of three parts. In first part, respondents were asked about their basic information on gender, age, weight, height and occupation. The second part asked about respondents' locations and activities. Respondents were asked to report whether they were in an indoor or outdoor environment, or in a vehicle, and those who reported that they were indoor were then asked about specific type of building in which they were staying. Two questions asked about their current activity and major activity in the one hour before, with options of seated, standing, slowly walking, fast walking, light exercise, intense exercise, sleeping and others. The last part asked about their sensations and satisfactions with the four environmental factors and the overall environment. As quality control check, a trap question was included in each questionnaire to identify responses which are not carefully made. The trap question had an obviously correct answer, for instance, what is the capital of China? The trap question was randomly extracted from a database for each questionnaire when it was generated and sent to respondents. The online survey obtained a total of 1869 samples, of which 45 samples were identified as invalid by quality control check and thus being excluded. Another 298 of samples were excluded as respondents reported they were not indoors. Thus, a number of 1526 valid samples were analyzed in the following sections.

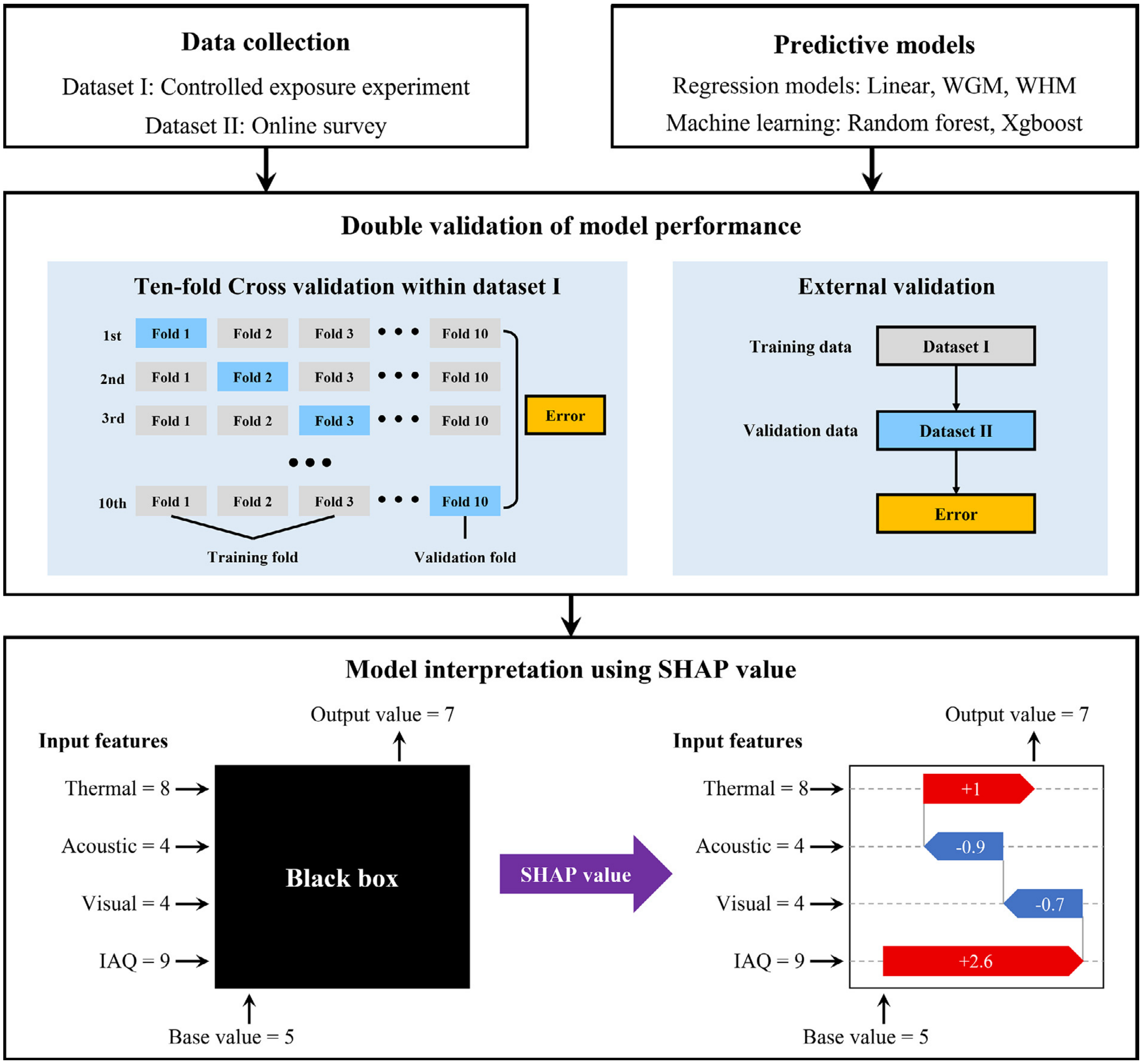

Fig. 1. Schematic outline of the methodology.

3.4. Distribution of IEQ satisfactions

The subjects were asked to rate their current satisfaction using a scale from 1 (very unsatisfied) to 10 (very satisfied) in the experiment I and the online survey, and using a scale from 0 (very unsatisfied) to 10 (very satisfied) in the experiment II. However, no subjects rated their satisfaction with a factor of IEQ or the overall IEQ at 0 throughout the experiment II. The example of question is shown below.

Please describe your satisfaction with the thermal/acoustic/visual/IAQ/overall environment using a rating from 1 (very unsatisfied) to 10 (very satisfied).

O 1 O 2 O 3 O 4 O 5 O 6 O 7 O 8 O 9 O 10

Fig. 2 illustrated the distribution of satisfactions with the IEQ factors and the overall IEQ in the two datasets. The satisfactions of the two datasets both showed a right skewed distribution, that the mean value was lower than the median. Sixty-five percent of the subjects/respondents rated their satisfactions between 7 and 9, while the most vote went to 8 (27%). Only 3% of subjects/respondents rated their satisfactions at 3 or lower. Thermal satisfaction had the highest mean at 7.46 and the overall satisfaction had the lowest at 7.08. ANOVA (Analysis of variance) revealed the significant difference in satisfactions between IEQ factors and the overall IEQ ($P < 0.001$). ANOVA also confirmed that the mean satisfaction of dataset II (7.39) was significantly higher than the dataset I (7.20) ($P < 0.001$). More respondents rated their satisfactions with IEQ factors and overall IEQ at 10 in dataset II than dataset I. More summary statistics of the two datasets can be found in Tables S3 and S4 in Supplementary Material.

4. Methodology

4.1. Linear and non-linear regression model

One linear and two non-linear regression models were built to predict the overall satisfaction (OS) with thermal satisfaction (TS), acoustic satisfaction (AS), visual satisfaction (VS) and IAQ satisfaction (IAQS).

The multivariate linear model was built using the Ordinary Least Squares (OLS) regression to minimize the sum of the squares of the difference between predicted value and observed value, defined as:

Table 2
Characteristics of the dataset I and dataset II.

Characteristic	Dataset I		Dataset II
	Experiment I	Experiment II	
Methodology	Controlled exposure and questionnaire		Online survey
Number of subjects/respondents	8	48	1869
Number of IEQ conditions	31	12	–
Number of valid responses	496	1088	1526
Occupation	University students		Students (14%), nonstudents (86%)
Age	20 to 25		16 to 72
Female to male ratio	50% to 50%		51% to 49%
Environment	Office		Office (44%), residential (28%), school (8%), hospital (1%), shopping mall (2%), hotel (1%), outdoor & vehicle (16%)

Table 3
Simulated parameters and levels in the controlled exposure experiments.

Experiment	N of IEQ conditions	Simulated parameters	Designated levels
Experiment I	31	PMV	−1.5, −0.97, −0.5, 0, 0.5, 0.97, 1.5
		Sound pressure (dBA)	40, 44.7, 48.5, 52.5, 56.5, 60.6, 65
		Illuminance (lx)	100, 259, 400, 550, 700, 841, 1000
		CO_2 (ppm)	500, 677, 830, 1000, 1160, 1323, 1500
Experiment II	12	Temperature (°C)	25, 27, 29
		Sound pressure (dBA)	40, 45, 50
		Illuminance (lx)	150, 300, 500
		Ventilation rate per person (m^3/h)	0, 30, 60

$$\widehat{OS} = \beta_0 + \beta_1 \times TS + \beta_2 \times AS + \beta_3 \times VS + \beta_4 \times IAQS \tag{1}$$

As the linear model can be regarded as a weighted arithmetic mean of predictor variables, we attempted to draw on other statistical indicators to build nonlinear regression models. Three classical Pythagorean means – arithmetic mean, geometric mean, and harmonic mean – of the four single-domain satisfactions were calculated and compared with the observed overall satisfaction. Three metrics – RMSE (Root Mean Square Error), MAE (Mean Absolute Error) and MAPE (Mean Absolute Percentage Error) – were used to measure the distance between the three Pythagorean means and the overall satisfaction, as shown in Table 4. The smaller the RMSE, MAE and MAPE, the more accurate the result.

All the metrics indicated that the overall satisfaction was closer to the geometric mean and harmonic mean of the four single-domain satisfactions than the arithmetic mean. The results indicated that the other two types of mean may better reflect the relationship between the single-domain satisfactions and the overall satisfaction. Thus, two nonlinear models were proposed based on the two means, namely weighted geometric mean (WGM) model and weighted harmonic mean (WHM) model.

The WGM model was mathematically defined as:

$$\widehat{OS} = \beta_0 + \sqrt[4]{TS^{\beta_1} \times AS^{\beta_2} \times VS^{\beta_3} \times IAQS^{\beta_4}} \tag{2}$$

The WHM model was mathematically defined as:

$$\widehat{OS} = \beta_0 + \frac{4}{\frac{\beta_1}{TS} + \frac{\beta_2}{AS} + \frac{\beta_3}{VS} + \frac{\beta_4}{IAQS}} \tag{3}$$

The WGM and WHM model were built using Nonlinear Least Squares (NLS) regression where a list of starting estimates needed to be designated to the parameters and then the parameters were refined by successive iterations. We set the starting estimates of all parameters at one, which was found to be an appropriate configuration as the models successfully converged.

It is noted that the number of parameters was intentionally kept equal between the three regression models, that each model has five parameters including an intercept. Thus, the parameters in the three models had the same degrees of freedom (df = 1535), which means the parameters had the same space to vary and the models had the same flexibility [34]. To resist overfitting, we controlled the complexity of the proposed regression models by not including interactions between the predictor variables. For the same reason, polynomial model using automatic procedure for predictive variable selection, for instance, stepwise regression, was not used in this study as it has been criticized of being easily overfitted and not replicable [35,36].

4.2. Machine learning model

Machine learning (ML) has been increasingly employed in prediction of the IEQ, building energy assumption and other building performance [22,37,38]. The major advantage of ML model is that it can automatically identify and learn underlying patterns in massive data and almost no human intervention is required for specifying the structure of the model. With the complex model structure built based on extensive computation, the ML model was found to be of greater predictive power than statistical model while it is also facing the barrier of interpretability and generalization [39]. Meanwhile, performance of a ML model heavily relies on settings of hyperparameters that define the model architecture. To build an optimal ML model, the tuning of model hyperparameters is crucial. In this study, we demonstrated building, tuning and verification of two popular tree-based ML models – Random Forest (RF) and Extreme Gradient Boost (Xgboost) on the obtained data. Due to its computational high efficiency, flexibility and portability, the RF and Xgboost have been widely applied in building related studies such as environmental issue identification [23,40–42], IEQ perception prediction [24,43], occupancy prediction [37,44,83], occupant behavior analysis [45,82], anomaly detection [46] and building energy analysis [47–49].

4.2.1. Random forest

A RF consists of a collection of classification and regression tree (CART), and the majority or average of predictions of all the CART become the prediction of the random forest [50]. A CART is built by iteratively dividing source data into subsets and finding best

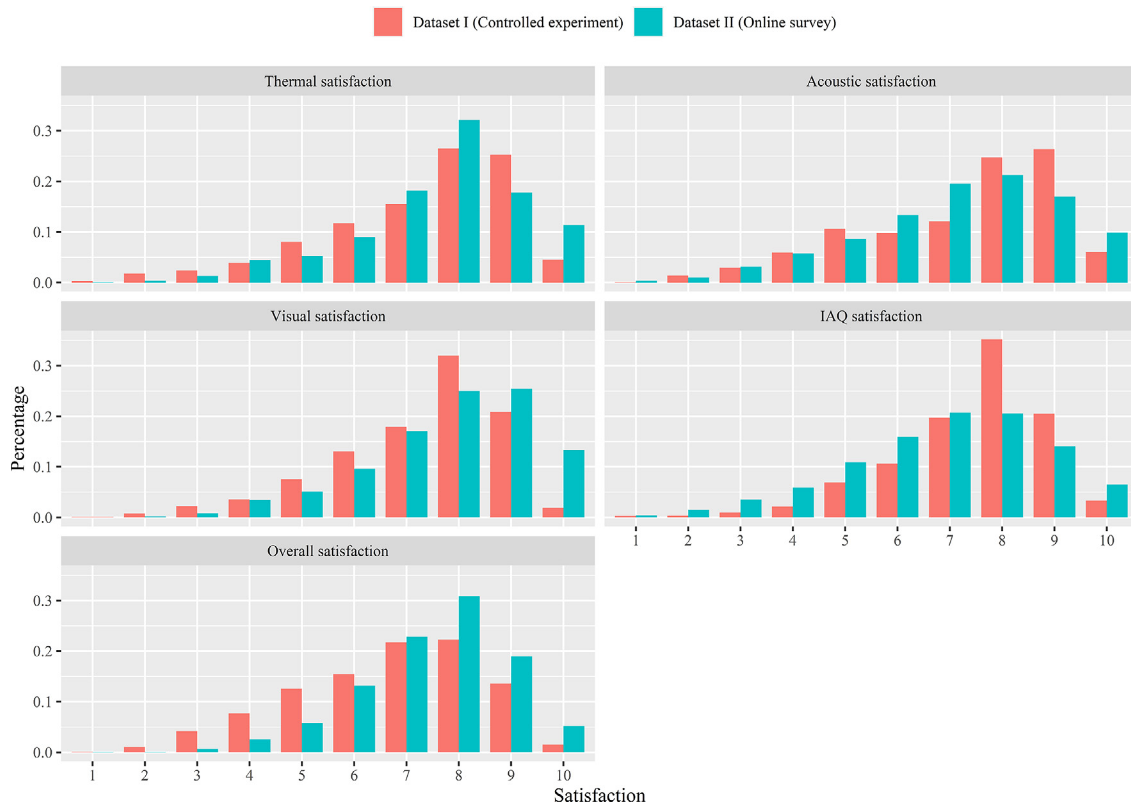

Fig. 2. Distribution of single-domain satisfactions and overall satisfactions in dataset I and dataset II.

Table 4
Using different metrics to measure distance between the overall satisfaction and means of the four single-domain satisfactions.

Metrics	Means of the four single-domain satisfactions		
	Arithmetic mean	Geometric mean	Harmonic mean
RMSE	1.15	1.04	0.97
MAE	0.81	0.76	0.72
MAPE	0.16	0.14	0.13

split at each node to reduce the impurity or variance in subsets as much as possible. In this process, depending on type of target variable, different criterions could be used to evaluate the quality of split and look for the best predictor. Here we used the most common criterion for regression analysis – the sum of the squared error (SSE), and the loss function is defined as:

$$SSE = \sum_{i \in D_L(A,s)} (y_i - c_L)^2 + \sum_{i \in D_R(A,s)} (y_i - c_R)^2 \tag{4}$$

s, the node; A, the predictor used in the node; D_L, left subset of the node; D_R, right subset of the node; y_i, target value; C_L, mean of target value in left subset; C_R, mean of target value in right subset.

A fully developed CART is prone to overfitting as the split is repeated on each subset until the terminal nodes contain very less target values (five for regression by default) and the tree could be overly complex. Thus, using decision trees in an ensemble like random forest is encouraged to obtain a better predictive performance and resistance to overfitting. The essential mechanism of random forest is that the individual trees are independently trained by: 1. Using bootstrap sample of the data when growing a tree; 2. Each node is split using a random subset of input predictors, which is called feature bagging [51].

We tuned two hyperparameters when training the RF. The first hyperparameter mtry, which determines the number of features could be used to find the best split in each node, was optimized using a 10-fold cross validation (CV, introduced in subsection 4.3) and the best value determined by RMSE was two. The other hyperparameter ntree is the number of individual trees in RF. Generally, error of a RF reduces as the number of trees grows before the threshold from which increasing the number of trees would bring no significant performance gain [52]. In this study, we set the ntree at 1000 as it was far over the threshold and the time needed to train was around five min on an Intel Core i5-8350U CPU with 16 GB RAM.

4.2.2. Extreme gradient boost (Xgboost)

Extreme Gradient Boost, also termed as Xgboost, is a tree boosting algorithm proposed by Chen and Guestrin [53]. Different from training trees independently in RF, the subsequent tree in Xgboost was trained using the residuals of the preceding tree, and this process is called gradient boost. The Xgboost is thus constructed by sequentially adding new trees to address the variations which the preceding trees are unable to explain. To resist overfitting, Chen and Guestrin designed an essential mechanism for Xgboost by including a regularization term Ω in its loss function to penalize the complexity of the model, which is defined as:

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \|w\|^2 \tag{5}$$

Where T is the number of leaves in the tree, w is the leaf weights.

And the objective of minimization is defined as:

$$\mathcal{L}(\phi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_x) \tag{6}$$

Where $\hat{y}_i$ is the predicted value, y_i, is the observed value, l is a differentiable convex loss function defined in reference [53].

Four hyperparameters in Xgboost were tuned using grid search and 10-fold cross validation (CV) to build an optimal Xgboost

Table 5
Grid search for tuning hyperparameters used in Xgboost.

Hyperparameters	Description	Search range	Optimistic value
max_depth	Maximum depth of a tree. A greater value will make the model more complex and more likely to overfitting.	[4:8], integer	5
eta	Step size shrinkage used in update to prevents overfitting.	[0.01, 0.2]	0.05
subsample	Ratio of the training data used to grow a tree. A lower subsample will help to avoid overfitting.	[0.6, 0.9]	0.733
nround	Max number of boosting iterations (trees).	[1, 1000], integer	107

model. The description and search range of the hyperparameters were presented in Table 5. In this process, a total of 1000 Xgboost models were built using hyperparameters randomly selected in search ranges and the best set of hyperparameters were located by finding the model with the lowest mean RMSE (Root Mean Squared Error) on test data in the 10-fold CV. The best set of hyperparameters was given in Table 5.

4.3. Double validation of model performance

Generalization capacity is a common concern for predictive models especially those trained using ML approach [54]. To assuage concerns with overfitting, a general solution is to train models using part of the entire data and verify it using the remaining [55]. In this study, we demonstrated a double validation to evaluate performance of resulting regression models and ML models.

First, the prediction accuracy of all resulting models was evaluated using a 10-fold cross validation (CV) within dataset I [56]. In this process, the dataset I was randomly partitioned into 10 equal size subsamples. Each of the 10 subsamples was used as test data once and the remaining subsamples were used as training data. The process repeated 10 times so that the model could be verified on the entire data. Using this approach, CV is able to provide more robust predictive evaluation than the non-exhaustive methods where generally 20% or 30% of the data are used for verification [57,58]. The 10-fold CV was also used to tune hyperparameters for RF and Xgboost.

Then, an external validation was conducted where the models trained using the entire dataset I with tuned hyperparameters were tested on dataset II. Since CV, as an internal validation approach, cannot guarantee the quality of a predictive model due to potentially biased training data [54], the external validation was applied to verify if resulting models were applicable to a dataset collected from a larger and more diverse populations from different building types.

The RMSE was used as the metric to measure difference between predicted and observed overall satisfaction for resulting models in the double validation.

4.4. Interpretation of models using sharply additive explanations (SHAP)

A major objective of this study is to interpret how occupants' overall satisfaction is determined by their satisfactions with individual environmental factors. However, predictions of data-driven models are often difficult to explain as it is constructed as a black box [39]. To enhance the interpretability, some ML algorithms provide metrics to evaluate contribution of input features, such as the mean decrease in accuracy (MDA) and the mean decrease in Gini (MDG) of RF [59]. Such metrics are always highly customized for certain ML algorithm and not applicable for others.

SHapley Additive exPlanations (SHAP) is an unified framework to provide comparable interpretation of predictive model with regard to the contribution of input features [60]. The objective of SHAP is to explain prediction of a complex model by attributing an effect to each feature which is measured by SHAP value. To compute SHAP value, a linear explanation model is used as an interpretable approximation to the complex original model, defined as:

$$g(z') = \phi_0 + \sum_{i=1}^{M} \phi_i z'_i$$

Where $z' \in \{0,1\}^M$, represents whether a feature is used ($z' = 1$) or not ($z' = 0$) to predict the target variable; M is the number of input features; ϕ_i is the SHAP value for the i^{th} feature; ϕ_0 is the base value, the mean of target variable in dataset.

Then, the effect of a feature is computed by comparing predictions of the model $f_{S\cup\{i\}}$ built with the feature present and the model f_S built with the feature withheld. Given the presence of multicollinearity between features, the differences are computed for all possible subsets of the set of all features F. The weighted average of all possible differences is the classic Shapley value (Eq. (8)), and the SHAP value is the Shapley value of a conditional expectation function of the original model [60].

$$\phi_i = \sum_{S\subseteq F\{\{i\}} \frac{|S|!(|F|-|S|-1)!}{|F|!}\left[f_{S\cup\{i\}}\left(x_{S\cup\{i\}}\right) - f_S(x_S)\right] \quad (8)$$

The SHAP values are intuitive and easy to understand. It can provide local explanation of quantitative contributions of features, while most importance metrics only evaluate effect of features on the entire dataset level. An illustration of SHAP is given in Fig. 3. For the given instance, the SHAP values assigned to a thermal satisfaction (TS) of 4 and a visual satisfaction (VS) of 9 were −1.26 and 0.63, respectively, indicating the TS reduced the overall satisfaction by 1.26 and the VS improved the overall satisfaction by 0.63, on a base value of 6.83. The difference between the output value f(x) and the base value of the overall satisfaction is equal to the sum of SHAP values assigned to all input features, which in this instance is −0.64.

As a unified framework, the SHAP was widely used to interpret complex ML method including the Xgboost [61,62] and RF [63,64], while the statistical models were less studied perhaps because of its transparency. In this study, we calculated the SHAP values for both regression models and ML models to interpret how predictions were made by different models and provide an evaluation of contribution of the four single-domain satisfactions on a comparable basis. The kernel explainer of SHAP was used to interpret regression models and the tree explainer was used to interpret RF and Xgboost.

4.5. Software and package

The OLS regression and NLS regression were performed in R with the base package "stats". The package "randomForest" [51] and "xgboost" [65] in R were used to train random forest and Xgboost, respectively. The "caret" [66] in R was used to perform CV and tune model hyperparameters. The SHAP values were calculated and visualized in Python using "shap" package [60,67].

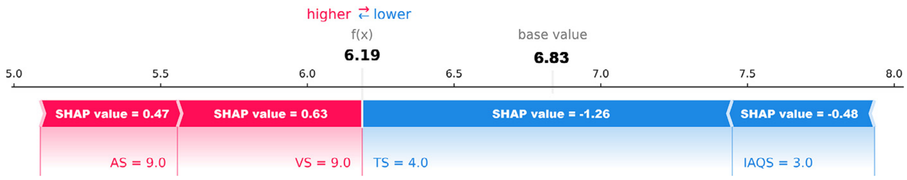

Fig. 3. An illustration of Shapley Additive Explanations, the red arrow represents positive impact of feature on target variable while blue arrow represents negative impact.

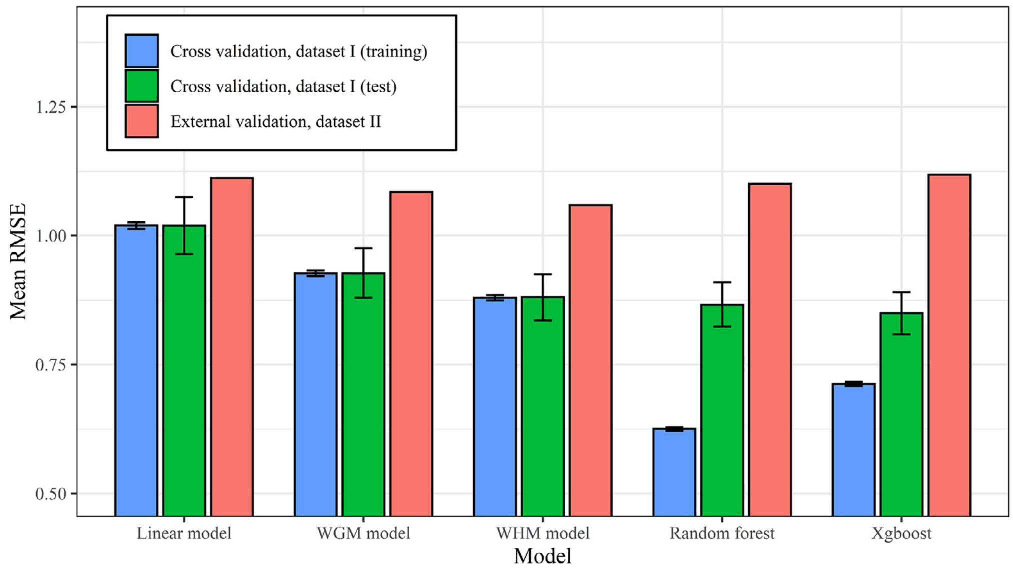

Fig. 4. Prediction error of resulting models in the 10-fold CV and external validation.

5. Results

5.1. Evaluation of model accuracy and applicability

The mean RMSE of established models on training and test data within dataset I estimated by 10-fold CV is presented in Fig. 4. The error bar represents standard deviation (SD). Fig. 4 also presented the prediction error of models in the external validation where they were trained using entire dataset I and tested on dataset II.

The nonlinear regression models showed a better accuracy on test data than the linear model in CV, that the mean RMSE on test data was 1.02 for linear model, 0.93 for WGM model and 0.88 for WHM model. For the three regression models, the mean accuracy on training and test data was almost the same, indicating a good resistance to overfitting. However, the increased SD of RMSE on test data indicated a less robustness of predictions. The accuracy of ML models was slightly higher than the regression models on test data, that the mean RMSE was 0.87 for RF and 0.85 for xgboost. Meanwhile, there is evidence of overfitting that the accuracy of the two ML models was significantly higher on training data than on test data.

As expected, the prediction error of resulting models increased when they were applied to dataset II, which was possibly due to the greater variance in respondents, indoor environment conditions and other potential confounding factors. The average percentage of increased error was 21% for the five models. The prediction error increased most for the two ML models by 29% while the linear model had the slightest increment by 10%. This is partly because the linear model has the simplest model structure which makes it less sensitive to noisy data and easier to generalize. The WHM model was found to be the most accurate model on dataset II with a RMSE of 1.05, followed by the WGM model with a RMSE of 1.09. Overall, the regression models showed a better applicability than the ML models.

5.2. Model performance on different building types and populations

To further evaluate the applicability of proposed models, the resulting models were tested on subgroups of dataset II divided by building types, respondents' age ranges and gender.

Fig. 5 showed the prediction error of models for different building types. The label represented number of samples collected from each type of building, and the dash line represented the mean RMSE of the model for entire dataset II. The distribution of prediction error for different building types was similar across the models. Except the hospitals with only 17 samples, office buildings had the lowest prediction error. However, the mean error for office buildings was still higher as compared to the mean error on the training dataset – dataset I. The prediction error increased more significantly when predicting overall satisfaction for residences and schools, and the error was above the average on dataset II. For the remaining building types, the prediction error was low for hospital while increased prediction error was found for shopping mall, hotel and other building types. However, few of samples was collected from those building types and thus we could draw no firm conclusion about the applicability of models to those building types.

Since the dataset I (training data) was collected from simulated office environment, it was not surprised to find that the model fitted better to office buildings than others. To clarify which factors led to the different perceptions of office, school and residential environment, three WHM models were built on samples from office, residence and school in dataset II for comparison, as shown in Table 6. The WHM model was applied here as it was found to be

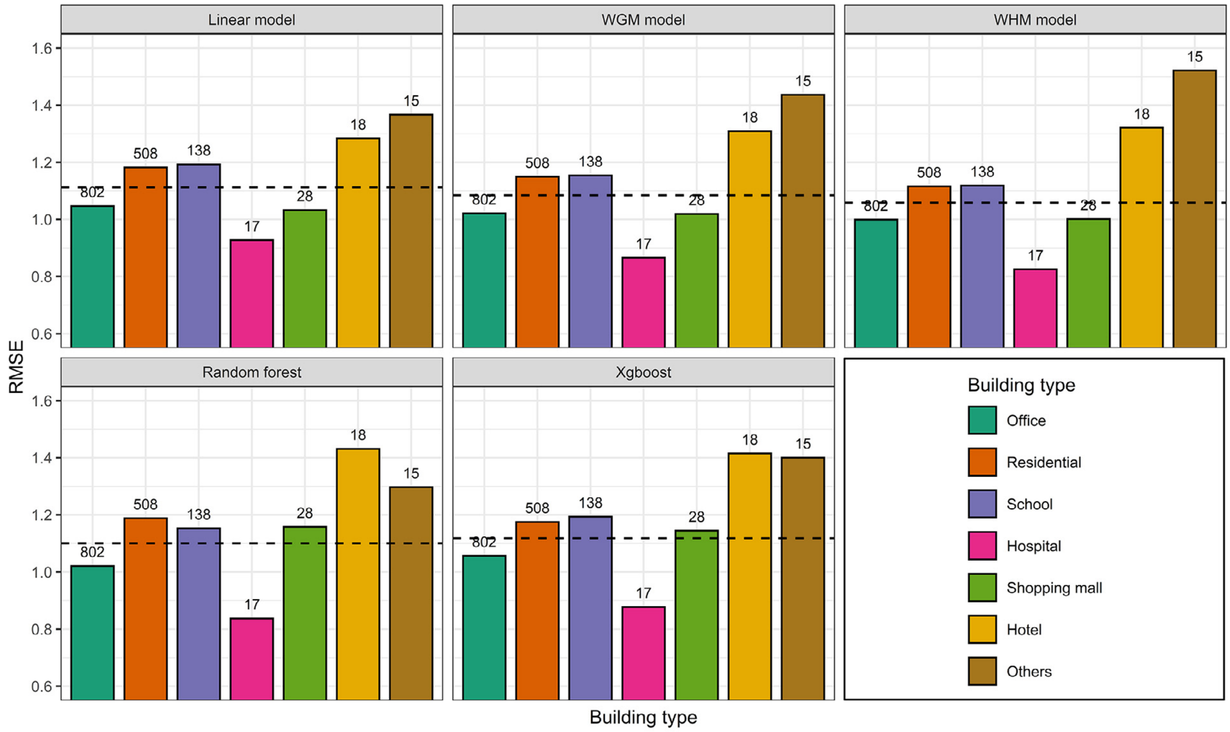

Fig. 5. RMSE of models on subgroups of dataset II divided by building types, the labels represented number of respondents, the dash line represented mean RMSE on entire dataset II.

Table 6
Coefficients of WHM model built on samples from offices, residences and schools.

Single-domain satisfaction	Coefficients of WHM model		
	Office	Residential	School
TS	1.14***	1.13***	0.93***
AS	0.84***	0.90***	1.13***
VS	0.99***	1.01***	0.97***
IAQS	1.05***	1.02***	1.02***

Note: *, $P \leq 0.05$; **, $P \leq 0.01$; ***, $P \leq 0.001$.

the most accurate model for dataset II. A higher coefficient in model indicates greater influence of a predictor. The result showed that the acoustic satisfaction was considered more influential by respondents in residence and schools than respondents in offices, and it was considered as the most influential factor in schools. Thermal satisfaction was found to have approximately equal importance in offices and residentials, but lower importance in schools. Similar coefficient for visual and IAQ satisfaction was found for three building types. All the coefficients reached significance level ($P < 0.001$).

The RMSE of resulting models on subgroups of dataset II divided by age ranges was presented in Fig. 6. The prediction error was higher for respondents with age not greater than 20 than other ranges. A major reason is that a considerable portion of the young respondents was also the student respondents in school, and the percent was 90% for the subgroup with age not greater than 20 and 15% for the subgroup with age between 20 and 30. This result was consistent with the finding that the model showed a worse performance when predicting overall satisfaction in school (Fig. 5). The prediction errors were close on subgroups with age ranges of 20 to 30, 30 to 40 and 40 to 50, and were close to the average error on the entire dataset II. The prediction errors were lower for subgroups aged greater than 50, but this result needs further verification due to the limited number of samples falling within this range.

The prediction error of models was slightly lower for male respondents (1.05) than female ones (1.14), as shown in Fig. 7. The result was of high consistency across the models. By comparing coefficients of WHM model fitted to two gender groups (Table 7), we found that female respondents' overall satisfaction was more often influenced by their acoustic satisfaction compared to males, while males considered IAQ more important. Coefficients for thermal and visual environment were similar between the two gender groups.

5.3. Interpretation of model predictions using SHAP

SHAP values were calculated to interpret attribution of each single-domain satisfaction to the overall satisfaction. The summary SHAP plot illustrated the distribution of SHAP values of input features in the five models, as shown in Fig. 8(a–e). The horizontal coordinate reflects the SHAP values, while the predictors were presented on vertical coordinate ordered by their mean absolute SHAP values. Each dot represents a sample from dataset, and the gradient color represents the value of corresponding predictor from low (blue) to high (red). As introduced in subsection 4.4, the negative and positive sign of SHAP value indicates the direction of feature impact, and the absolute SHAP value describe the amount of feature impact. Since all the single-domain satisfactions were positively associated with the overall satisfaction, there is a clear trend that a dot with gradient color close to red is located on the right side, indicating a positive impact, while dot with more blue located on the left side indicating a negative impact. One will note that the distribution of SHAP values of linear model showed a discrete pattern with equal interval, which reflects the absolute proportional relationship between the single-domains satisfaction and overall satisfaction.

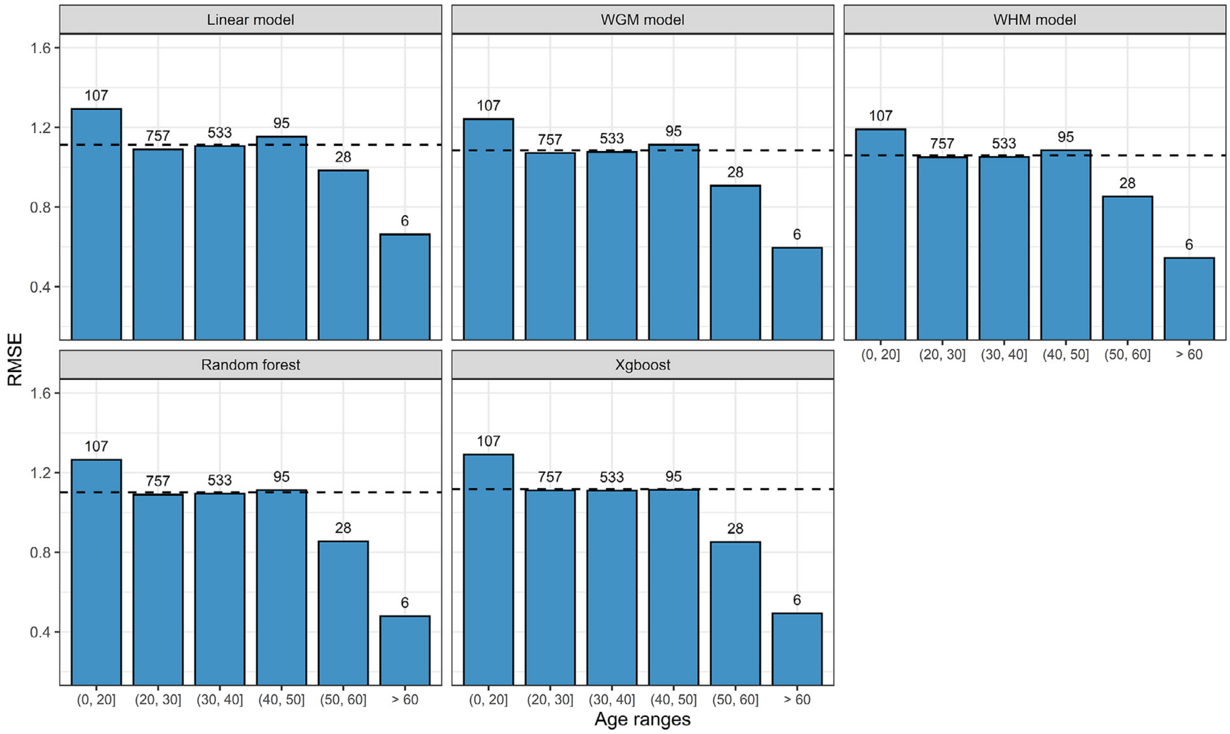

Fig. 6. RMSE of models on subgroups of dataset II divided by age ranges, the labels represented number of respondents, the dash line represented mean RMSE.

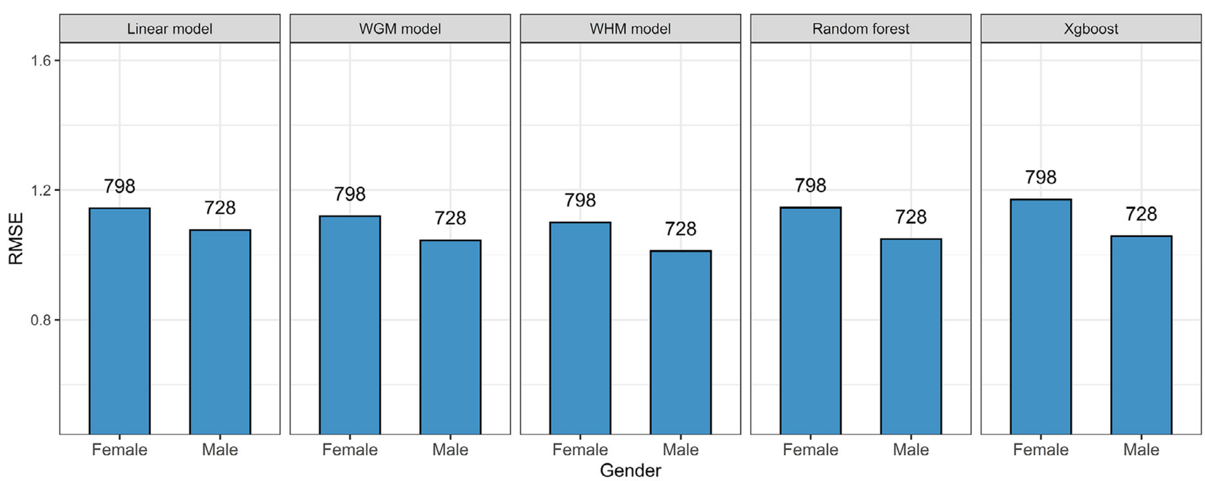

Fig. 7. RMSE of models on subgroups of dataset II divided by gender, the labels represented number of respondents.

Table 7
Coefficients of WHM model built on female and male groups.

Single-domain satisfaction	Coefficients of WHM model	
	Female	Male
TS	1.12^{***}	1.09^{***}
AS	0.95^{***}	0.86^{***}
VS	0.98^{***}	1.02^{***}
IAQS	1.00^{***}	1.08^{***}

Note: *, $P \leq 0.05$; **, $P \leq 0.01$; ***, $P \leq 0.001$.

The summary plots revealed that, an environmental factor tended to have dominate negative impact on the overall satisfaction when it was unsatisfying, and this negative impact was more significant than the positive impact of the factor when it was satisfying. This inference was drawn from the pattern that the bule dots and red dots are asymmetrically distributed on both sides of the centerline, that the blue dots are located further from centerline and had a greater absolute SHAP value than those with red color. Besides, the negative impact of unsatisfying factors was more significant in non-linear models and ML models than in the linear model. For example, a low acoustic satisfaction could push down the overall satisfaction by more than five points in the WHM model, while it only caused about 2.5 points of decrease in the linear model.

The dependence SHAP plot further illustrated the nonlinear relationship between single-domain satisfactions and the overall satisfaction. Fig. 9 presented the SHAP values for the thermal satisfaction as a function of its value in the linear model, WHM model and Xgboost. There is an obvious trend in SHAP values of WHM and

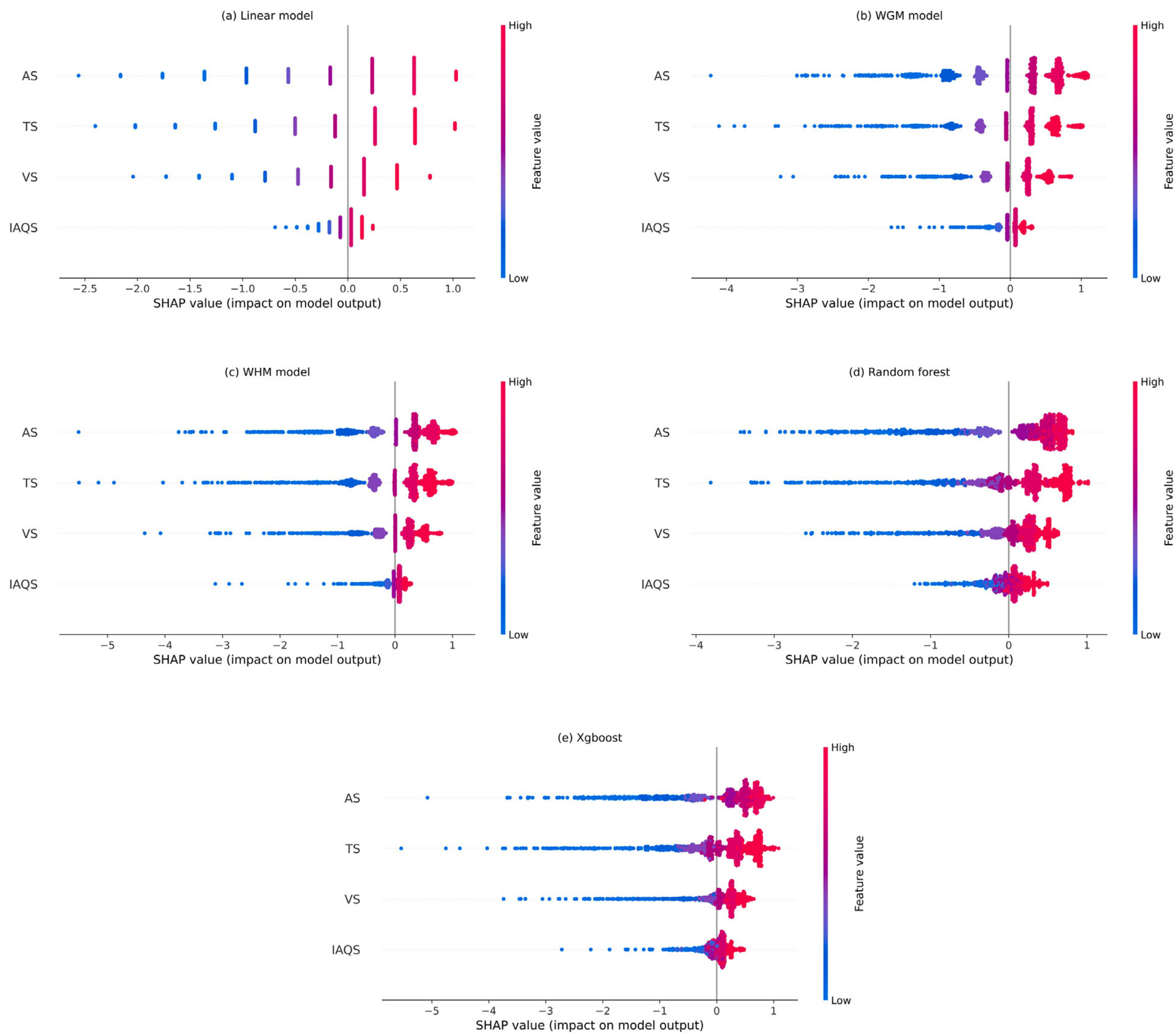

Fig. 8. Distribution of SHAP values of single-domain satisfactions as input features in (a) Linear model (b) WGM model (c) WHM model (d) Random forest (e) Xgboost, the features were sorted by the mean absolute SHAP value, the color represents value of features from low (red) to high (blue).

Xgboost model that the negative impact of thermal satisfaction increased rapidly with thermal satisfaction reducing and the impact was much more influential as compared to the proportional relationship in linear model. The results showed that the overall satisfaction was more likely to be dragged down by a low single-domain satisfaction rather than be benefited from a high single-domain satisfaction along. The figure also showed the significant variance in SHAP values in nonlinear and ML models, indicating the presence of interactions between variables. For example, a thermal satisfaction of 4 had a constant negative impact of −1.3 on the overall satisfaction suggested by linear model, while its impact fluctuated from −0.9 to −1.8 in WHM model and from −0.4 to −2.7 in Xgboost, depending on the satisfaction level of other factors.

6. Discussion

6.1. Relationship between single-domain satisfactions and overall satisfaction

The main objective of this study was to interpret how occupants' overall satisfaction was determined by several single-domain satisfactions with physical environmental factors. While the linear regression model was employed in many of the previous studies (See Table 1), some studies have questioned the reliability of this linear relationship and have attempted to use other statistical methods to extend the understanding of the correlation between different environmental factors [18,19]. By comparing accuracy and applicability of models on obtained data, we found that the linear model was underperformed in predicting the overall satisfaction compared to proposed nonlinear regression models and ML models. The result indicated that it may not be appropriate to simplify the relationship between satisfactions with individual environmental factors and the overall environment to a linear relationship. The SHAP values further revealed the fact that an environmental factor which was particularly unsatisfying tended to have more significantly negative impact on the overall satisfaction. Consistent result was reported by Kim and de Dear in their study in which they found some IEQ factors had a predominately negative impact on the overall satisfaction when it fails to meet occupant expectations [19].

The interpretation of relationship between single-domain satisfactions and the overall satisfaction enables building manager to establish effective strategy to improve IEQ by prioritizing factors

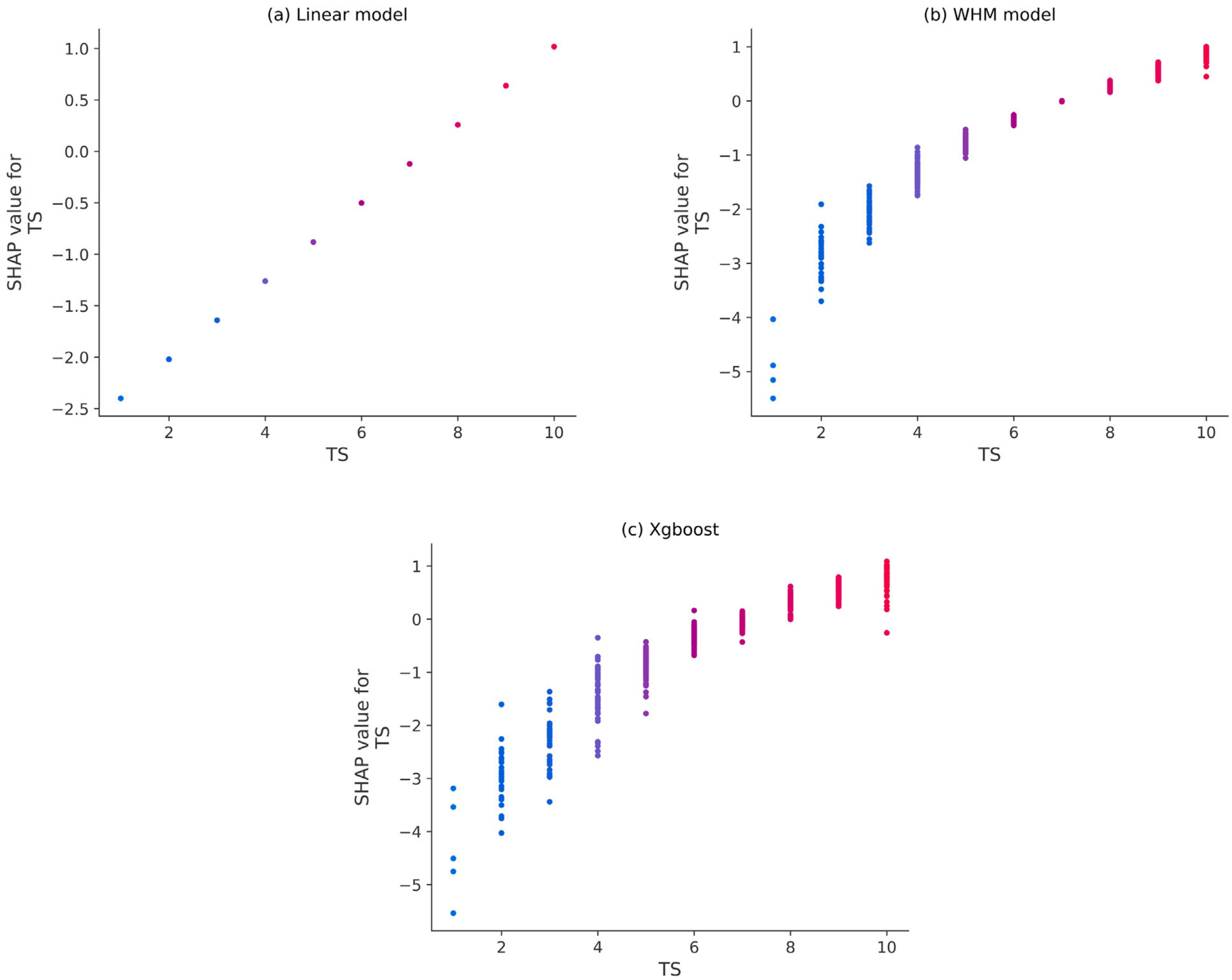

Fig. 9. Dependence plot of SHAP values of thermal satisfaction in (a) linear model (b) WHM model and (c) Xgboost model.

that have significant negative impact on the overall satisfaction. By estimating how much improvement in overall IEQ can be achieved by improving a factor to a certain level, the IEQ models can help building managers assess in advance the effectiveness of IEQ retrofit solutions. The gains in overall satisfaction could be significantly different based on the resource allocation for IEQ. For an instance where thermal satisfaction was 4 and the other three single-domain satisfactions were 8, increasing thermal satisfaction from 4 to 6 would increase the overall satisfaction by 1.23, while increasing visual satisfaction from 8 to 10 would only increase it by 0.25, suggested by the WHM model. The model suggested that it was more efficient to focus efforts and resources on the environmental factors which were not delivered at satisfactory levels rather than investing more into factors which are already considered satisfied [19]. At the same time, it is important to note that the difficulty of achieving high levels of satisfaction varies across environmental factors [20], which can also affect investment strategies. Future studies are needed to assess the difference in amount of energy, manpower and material resources required by improving perceptions of various environmental factors. Including an assessment of investment will enable the proposed IEQ models to provide additional insights to help building managers obtain more sustainable returns from operations management and IEQ retrofit.

6.2. Regression models and machine learning models

Although we perhaps underutilized the potential of ML algorithms, we found that the fine-tuned ML models did not substantially perform better than the statistical non-linear models in predicting overall satisfaction. CV within dataset I (training data) showed that the accuracy of RF and Xgboost has a small advantage over the proposed regression models, while the validation on dataset II indicated that the regression models had better generalizability than the ML models. This may be because that the strength of ML method lies in addressing high-dimensional problems involving high number of features, inapparent patterns in data and complex interactions between features [68,69]. When only four environmental factors were considered, the non-linear regression models, such as the proposed WHM model, are capable of exploring the relationships between predictors and achieve an accuracy close to that of the ML model.

Another advantage of the regression model is that it is more intuitive and easier to understand and use by the average building manager and technician as compared to ML models, considering the ML models always require larger datasets and sometimes the involvement of professionals for their establishment, tunning and interpretation [70,71]. It is recommended for building managers to build their own IEQ model based on a small-scale survey conducted in their buildings to minimize biases resulting from populations and environment. In cases where survey resources are limited, a simplified index derived from suitable regression model, such as a harmonic mean of single-domain satisfactions, can be used to roughly estimate overall satisfaction, despite the potential loss of accuracy. The transparent structure of regression models also makes it more likely to be adopted by a standard or guideline that aim to assess the overall IEQ of building based on the performance of single environmental aspects [6].

6.3. Model performance and subject samples

In studies involving human subjective perceptions, individual difference is a key factor that cannot be neglected when evaluating applicability of models [72]. Many of IEQ studies suffered from limited number and source of samples, and that potentially reduced applicability of their conclusions to more general scenario [23,73]. In this study, the applicability of proposed IEQ models were tested by comparing their accuracy when trained on data from controlled environment and when applied to data from real buildings. Increased error was found in the validation especially for the ML models. This can be interpreted as that the better the ML models learned patterns from a limited dataset, the less generalization it could achieve [74]. The models performed better when they were applied to the data collected from the same type of environment as the training data, which in this case is office buildings. The various activities of the occupants have given rise to different perceptions of IEQ, which determines their various demands for different aspects of indoor environment. The respondents in schools were found to be more sensitive to the level of acoustic satisfaction than respondents from office. A possible reason for this is that the schoolwork requires quieter environment than normal office work. However, this cannot be verified in this study. There are still many issues that need to be further addressed regarding the differences in demands of occupants on IEQ, which is of great significance in customizing indoor environment that better satisfy occupants in different buildings.

Gender differences were found in assessment of overall IEQ in this study. Females were more sensitive to the quality of the acoustic and thermal environment than males, while the visual environment and IAQ had greater impacts on males' perception of overall IEQ than females. It has been reported in previous studies that females were more likely to be dissatisfied with IEQ factors than males [29,75]. However, in the present study, no significant difference was found in satisfaction with the four factors or the overall IEQ between genders. The differences in metabolic rate, clothing and activity between females and males were considered as possible reasons for different IEQ perception [76].

6.4. Importance of environmental factors

Many studies had reviewed the proposed IEQ models in the literature and noted the significant inconsistency with regard to relative importance of environmental factors specified by the models [8,9,21]. Since different approaches and predictor variables were used, the results were not directly comparable across models. For example, in a model considering three environmental factors, the factors are given inherently more weight than in a model that includes four or more factors. Therefore, the relative importance of factors specified by the proposed IEQ models were represented by their normalized coefficients, which were calculated by dividing the coefficients by the sum of coefficients attributed to the four factors, as shown in Fig. 10. The priority of variables is represented using numbers from one to four indicating most influential to most uninfluential based on the normalized coefficients. Five of the listed studies [8,12,13,15,20] evaluated all four principle IEQ factors, while the remaining [10,16,18,21] evaluated only three factors. It is noted that there could be information loss during the normalization, and there is uncertainty involved when comparing the coefficients estimated by different models.

There is a lack of consistency in terms of the relative importance of variables in IEQ models proposed by reviewed studies (Fig. 10). For example, despite having the highest average priority of 1.9, the

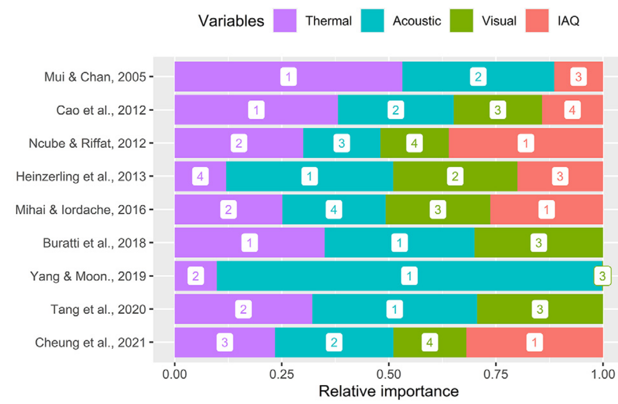

Fig. 10. Relative importance of environmental factors with regard to the impact on overall satisfaction, the number represents rank of the importance.

acoustics was still considered the least important and second least important in two studies. There is also a significant variance in importance of acoustic in studies which considered it most important. The acoustics was reported to dominate the overall satisfaction in the study by Yang & Moon [21] while it had a close importance with thermal satisfaction in studies by Buratti et al. [16] and Tang et al. [18]. The thermal and IAQ had an average priority of 2 and 2.17 respectively, but with significant variance across studies as well. The only consensus among the studies is that visual satisfaction is less influential with an average priority of 3.12 and small variance. Since different metrics were used to evaluate the accuracy of the proposed IEQ models, there is no comparability of the model performance across the reviewed studies.

However, the results of this study indicated that it may not be accurate to use a fixed scheme to represent the importance or priority of an environmental factor when assessing the overall performance of IEQ. This is not only because the occupants in different buildings were observed to value environmental factors differently, reflecting their various demands, but also because that occupants were likely to rate the most unsatisfactory factor as the most important one [7]. For example, if respondents more often perceived thermal environment as dissatisfied compared to others, such imbalance in data would yield a result that respondents rated thermal environment more important than others. For this reason, although the SHAP value indicated that the acoustic satisfaction contributed most to the overall satisfaction followed by the thermal, visual and IAQ in this study, we considered this to be a conclusion limited to the study populations and then-current environment. The finding of this study indicated that it is not rigorous to claim that a certain environmental factor is more important than the other as dissatisfaction with each of them could lead to low overall satisfaction which cannot be compensated for by a higher satisfaction with another factor. Rather than using a single metric to rank IEQ factors, we suggest that the evaluation of importance or priority of IEQ factors should be discussed in a specific context, taking into account the current IEQ conditions and populations.

6.5. Limitations of the study

Besides the four principle environmental factors investigated in this study, there are other contextual factors which could have significant impact on occupants' overall satisfaction, such as layout, cleanliness, privacy, window view, etc. The reason for not considering these factors is that they are more subjective, hard to measure, and difficult to control and simulate in laboratory. However, these factors were found to be influential and even have dominant impact on environmental perceptions under some scenarios [20]. Higher accuracy could be achieved by including occupants' satisfaction with more contextual factors in predictive models. Meanwhile, it is possible that the accuracy of ML algorithms may be more advantageous than regression models on a high-dimensioned data where a large number of predictors are being considered [68]. Future work extending the range of environmental factors investigated is highly appreciated to provide additional insights into the interpretation of occupants' overall satisfaction with IEQ.

Another limitation is that insufficient samples were collected from hotel, shopping mall, hospital and other building types. This limitation prevented us from verifying the applicability of the proposed models in those types of building. There is also an imbalance in the age of respondents. The majority of respondents were between 20 and 40, and the lack of elderly respondents could be a problem when generalize the models to indoor environment designed for elderly occupants [77]. An important work in future is to evaluate the performance of the proposed IEQ model in a wider variety of environmental settings and populations.

At the methodology level, correlated variables are a tricky issue for all variable importance methods [78,79]. The importance of correlated variables may be splitting between correlated variables. Removing one of correlated variables often increases the importance of other variables, and biases in the training and test dataset can also change the importance rank, which leads to instability of variable importance. In the present study, the kernel SHAP explainer and tree SHAP explainer were used to minimize the effect of correlation between variables. However, the effect of correlated variables cannot be completely eliminated and may have a potential impact on results.

7. Conclusions

This paper outlined the establishment of five predictive models including regression models and ML models with an aim of achieving higher accuracy and interpretability in predicting overall satisfaction using four single-domain satisfactions – thermal, acoustic, visual and IAQ. The models were tested on two different datasets collected from controlled exposure experiments and existing buildings. The SHAP values were calculated to measure contributions of single-domain satisfactions to overall satisfaction for resulting models to provide the interpretation of relationship between them. The results of this study answered the following questions.

1. **What is the appropriate model for explaining relationship between occupants' satisfaction with individual environmental factors and the overall environment?**

Despite being used in many previous studies, the linear model was found to be underperformed. Lower accuracy for linear model was reported as compared to the proposed non-linear models and ML models, indicated by the cross validation within dataset I and the external validation on dataset II. The ML models had a slightly higher accuracy in cross validation than the proposed nonlinear regression models, while its error increased more significantly when applied to additional data. The small number of variables considered in this study could be a reason for the underperformance of the ML models. It is possible that the accuracy of ML models may further improve when additional contextual IEQ factors were taken into account. However, in practice, the accuracy should not be the only criterion for model selection. The transparency, interpretability and technical barriers of model should also be considered when evaluating the applicability of models especially for non-specialist building managers.

2. **When an IEQ assessment model fitted to a dataset is applied to additional data collected from different populations and environment, how much would the error increase and what factors contribute to the error?**

In this study, we trained the models on dataset I which was collected from recruited subjects in controlled exposure to simulated office environment, and then tested it on dataset II which was collected from respondents in various existing buildings. We observed that the prediction error of the models averagely increased by 21% on dataset II as compared to its performance in cross validation within dataset I. The prediction error of ML models increased more (by 29%) than regression models (16%), indicating a worse generalization of ML models due to its complexity. A further analysis on model performance on dataset II indicated that models were more generalizable to the samples collected from office buildings, which had similar settings with the training data, than schools and residences. Some differences in perceptions were identified across building types. For example, the quality of acoustic environment had more impact on respondents' overall satisfaction with IEQ in schools than that in office. The models performed not well among respondents under the age of 20, as most of them were also the respondents in schools. The models better predicted the overall satisfaction for males as compared to females. Females cared more about acoustic than males, while males cared more about IAQ.

3. **What inferences can be drawn from the established models and how these inferences will affect the management of IEQ?**

Using SHAP value, we explored how the single-domain satisfactions contribute to the overall satisfaction in the established models. An important finding is that an unsatisfactory environmental factor often had a dominate negative impact on occupants' perception that cannot be counteracted by higher satisfactions with other factors. As a consequence, in most circumstances the overall satisfaction benefited the most from the improvement of the least satisfied IEQ factor, and the benefits of improving a factor decreased with the level of its satisfaction. This finding is very different from the underlying assumption of the linear model that a constant amount of improvement in overall satisfaction can be achieved by improving a certain environmental factor. The nonlinear relationship highlighted the significance of resource allocation in IEQ management to develop an efficient path to achieve better IEQ overall performance. In the future, more studies are needed to extend the current IEQ models with contextual factors and consideration of investment to develop a useful toolkit for IEQ management and retrofit.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.enbuild.2022.111870.

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